USN | | | | | | | | | | [23CSE307]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

INTRODUCTION TO DATA SCIENCE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Define Data Science. How is it different from Computer Science.	L1	CO1	**02**
	b.	Infer the need of Feature Selection.	L2	CO2	**02**
	c.	Recognize five popular websites for dataset.	L2	CO3	**02**
	d.	Give example of Bivariate analysis	L1	CO4	**02**
	e.	Define sentiment analysis.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Recognize the implications of duplicate values in a dataset.	L2	CO1	**05**
		OR			
	b.	Illustrate a typical example of a dataset.	L2	CO1	**05**
3.	**a.**	Paraphrase methods for handling missing values.	L2	CO2	**05**
		OR			
	b.	Cite the handling methods for duplicate values.	L2	CO2	**05**
4.	**a.**	Elaborate Exploratory Data Analysis (EDA).	L3	CO3	**05**
		OR			
	b.	Describe Dimensionality Reduction.	L2	CO3	**05**
5.	**a.**	Describe stages of machine learning modeling cycle.	L3	CO4	**05**
		OR			
	b.	Define Machine Learning and its applications.	L1	CO4	**05**
6.	**a.**	Describe applications of social media analytics.	L2	CO5	**05**
		OR			
	b.	Describe applications of health care analytics.	L3	CO5	**05**
7.	**a.**	Elaborate Clustering with an example.	L3	CO4	**05**

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		OR			
	b.	Illustrate a dataset sample for regression analysis.	L3	CO4	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Describe the five points of data science process and the roles of data engineer, data analyst, machine learning engineer and data scientists.	L3	CO1	**10**
		OR			
	b.	Compare Problem Solving in Data Science vs Problem solving using Programming Languages (C, C++, Java).	L4	CO1	**10**
9.	**a.**	Summarize Data Transformation.	L4	CO2	**10**
		OR			
	b.	Interpret the Data reduction techniques.	L2	CO2	**10**
10.	**a.**	Name a few Data Visualization tools.	L2	CO3	**10**
	b.	Enumerate a few data science toolkit.	L3	CO3	**10**
11.	**a.**	Differentiate between Regression and Classification.	L2	CO4	**10**
		OR			
	b.	Differentiate between supervised and un-supervised machine learning.	L2	CO4	**10**
12.	**a.**	Paraphrase applications of data science in Business.	L2	CO5	**10**
		OR			
	b.	Describe basic prediction which a video recommender system does.	L3	CO5	**10**
13.	**a.**	Differentiate between machine learning and traditional programming.	L2	CO4	**10**
		OR			
	b.	Describe the relation between data science and machine learning.	L2	CO4	**10**